Vectors: Components and Operations

ANSWER KEY

Section 1: Geometric vs Component Form

1 Find the component form of vector $\mathbf{v}$ with initial point $P(2, -3)$ and terminal point $Q(-1, 4)$.

a. $\langle -3, 7 \rangle$ ✓

b. $\langle 1, 1 \rangle$

c. $\langle 3, -7 \rangle$

d. $\langle -2, -12 \rangle$

$\mathbf{v} = \langle x_2 - x_1, y_2 - y_1 \rangle = \langle -1 - 2, 4 - (-3) \rangle = \langle -3, 7 \rangle$.

2 Express $\mathbf{v} = \langle 5, -8 \rangle$ in terms of the standard unit basis vectors $\mathbf{i}$ and $\mathbf{j}$.

a. $5\mathbf{i} - 8\mathbf{j}$ ✓

b. $5\mathbf{i} + 8\mathbf{j}$

c. $-8\mathbf{i} + 5\mathbf{j}$

d. $5\mathbf{j} - 8\mathbf{i}$

$\mathbf{i} = \langle 1, 0 \rangle$ and $\mathbf{j} = \langle 0, 1 \rangle$, so $\langle 5, -8 \rangle = 5\mathbf{i} - 8\mathbf{j}$.

3 When are two geometric vectors considered equal?

a. **When they have both the same magnitude and the same direction** ✓

b. When they have the same initial point

c. When their lengths are equal regardless of direction

d. When they are collinear

Vectors are defined strictly by magnitude and direction, independent of their initial position in space.

Section 2: Magnitude and Direction Angle

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

4 Find the magnitude of $\mathbf{v} = \langle -6, 8 \rangle$.

a. 10 ✓

b. 14

c. $\sqrt{28}$

d. 2

$\|\mathbf{v}\| = \sqrt{(-6)^2 + 8^2} = \sqrt{36 + 64} = \sqrt{100} = 10$.

5 Find the direction angle $\theta \in [0^\circ, 360^\circ)$ of $\mathbf{v} = \langle -3, 3\sqrt{3} \rangle$.

a. 120° ✓

b. 60°

c. 150°

d. 240°

In Quadrant II: $\tan\theta = \frac{3\sqrt{3}}{-3} = -\sqrt{3}$. Reference angle is 60°, so $\theta = 180^\circ - 60^\circ = 120^\circ$.

6 Find the components of a vector with magnitude 12 and direction angle 210°.

a. $\langle -6\sqrt{3}, -6 \rangle$ ✓

b. $\langle -6, -6\sqrt{3} \rangle$

c. $\langle 6\sqrt{3}, 6 \rangle$

d. $\langle -6\sqrt{2}, -6\sqrt{2} \rangle$

$v_x = 12\cos(210^\circ) = 12(-\sqrt{3}/2) = -6\sqrt{3}$. $v_y = 12\sin(210^\circ) = 12(-1/2) = -6$.

Section 3: Vector Addition, Subtraction, and Scalar Multiplication

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

7 Given $\mathbf{u} = \langle 3, -2 \rangle$ and $\mathbf{v} = \langle -1, 5 \rangle$, find $2\mathbf{u} - 3\mathbf{v}$.

a. $\langle 9, -19 \rangle$ ✓

b. $\langle 3, 11 \rangle$

c. $\langle 9, 11 \rangle$

d. $\langle 3, -19 \rangle$

$2\mathbf{u} = \langle 6, -4 \rangle$. $3\mathbf{v} = \langle -3, 15 \rangle$. Difference $= \langle 6 - (-3), -4 - 15 \rangle = \langle 9, -19 \rangle$.

8 Find the unit vector $\mathbf{u}$ in the same direction as $\mathbf{v} = \langle 5, -12 \rangle$.

a. $\langle 5/13, -12/13 \rangle$ ✓

b. $\langle 5/17, -12/17 \rangle$

c. $\langle 1, -1 \rangle$

d. $\langle 5, -12 \rangle$

$\|\mathbf{v}\| = \sqrt{25 + 144} = 13$. Unit vector $\mathbf{u} = \frac{\mathbf{v}}{\|\mathbf{v}\|} = \langle 5/13, -12/13 \rangle$.

9 What is the result of multiplying any vector $\mathbf{v}$ by scalar -1?

a. **A vector of the same magnitude pointing in the exactly opposite direction** ✓

b. The zero vector

c. A scalar number

d. A perpendicular vector

Scalar multiplication by -1 negates every component, reversing the direction while preserving the magnitude.

Section 4: Resultant Forces and Navigation

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

10 Two forces act on an object: $\mathbf{F}_1 = \langle 20, 0 \rangle$ N and $\mathbf{F}_2 = \langle 0, 15 \rangle$ N. Find the magnitude of the resultant force.

a. 25 N ✓

b. 35 N

c. 5 N

d. 300 N

Resultant $\mathbf{F} = \langle 20, 15 \rangle$. Magnitude $\|\mathbf{F}\| = \sqrt{20^2 + 15^2} = \sqrt{400 + 225} = \sqrt{625} = 25$ N.

11 An airplane flies at a heading due east with airspeed 400 mph. A wind blows from the south at 50 mph. Find the ground speed of the plane.

a. $\sqrt{400^2 + 50^2} \approx 403.1$ mph ✓

b. 450 mph

c. 350 mph

d. 425 mph

Plane velocity $\mathbf{v}_p = \langle 400, 0 \rangle$. Wind velocity (blowing from south means heading north) $\mathbf{v}_w = \langle 0, 50 \rangle$. Total $\|\mathbf{v}\| = \sqrt{160000 + 2500} = \sqrt{162500} \approx 403.1$ mph.

12 In the airplane problem, what is the drift angle north of east?

a. $\arctan(50/400) \approx 7.1^\circ$ ✓

b. 12.5°

c. 45°

d. 3.6°

$\tan\theta = \frac{50}{400} = 0.125 \Rightarrow \theta = \arctan(0.125) \approx 7.1^\circ$.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647